

3(a) Using $pV = nRT$, [CONCEPT]

$$\begin{aligned}\text{Volume } V_f &= \frac{nRT}{p} \\ &= \frac{\text{Total mass}}{\text{Molar mass}} \frac{RT}{p} \\ &= \frac{350}{18} \times \frac{8.31 \times (273 + 100)}{1.0 \times 10^5} \\ &= 0.60271 \\ &\approx 0.60 \text{ m}^3\end{aligned}$$

(b) Work done, $W = p\Delta V$ [CONCEPT]

$$\begin{aligned}&= p(V_f - V_i) \\ &= p\left(V_f - \frac{m}{\rho}\right) \\ &= 1.0 \times 10^5 \left(0.6027 - \frac{0.350}{1000}\right) \\ &= 60235 \\ &\approx 6.0 \times 10^4 \text{ J}\end{aligned}$$

(Note: Volume of a gas at atmospheric pressure is so much larger than its volume as a liquid, so it's OK not to include V_i in the calculation.)

(c) $\Delta U = Q - W_{\text{by}}$ [CONCEPT]

$$\begin{aligned}&= mL_v - W_{\text{by}} \\ &= (0.350 \times 2.26 \times 10^6) - 60235 \\ &= 730770 \\ &\approx 7.3 \times 10^5 \text{ J}\end{aligned}$$

(d) The increase in internal energy takes the form of an increase in potential energy due to intermolecular forces of attraction.

4(a)(i) Since a and r are constant, so a is proportional to x